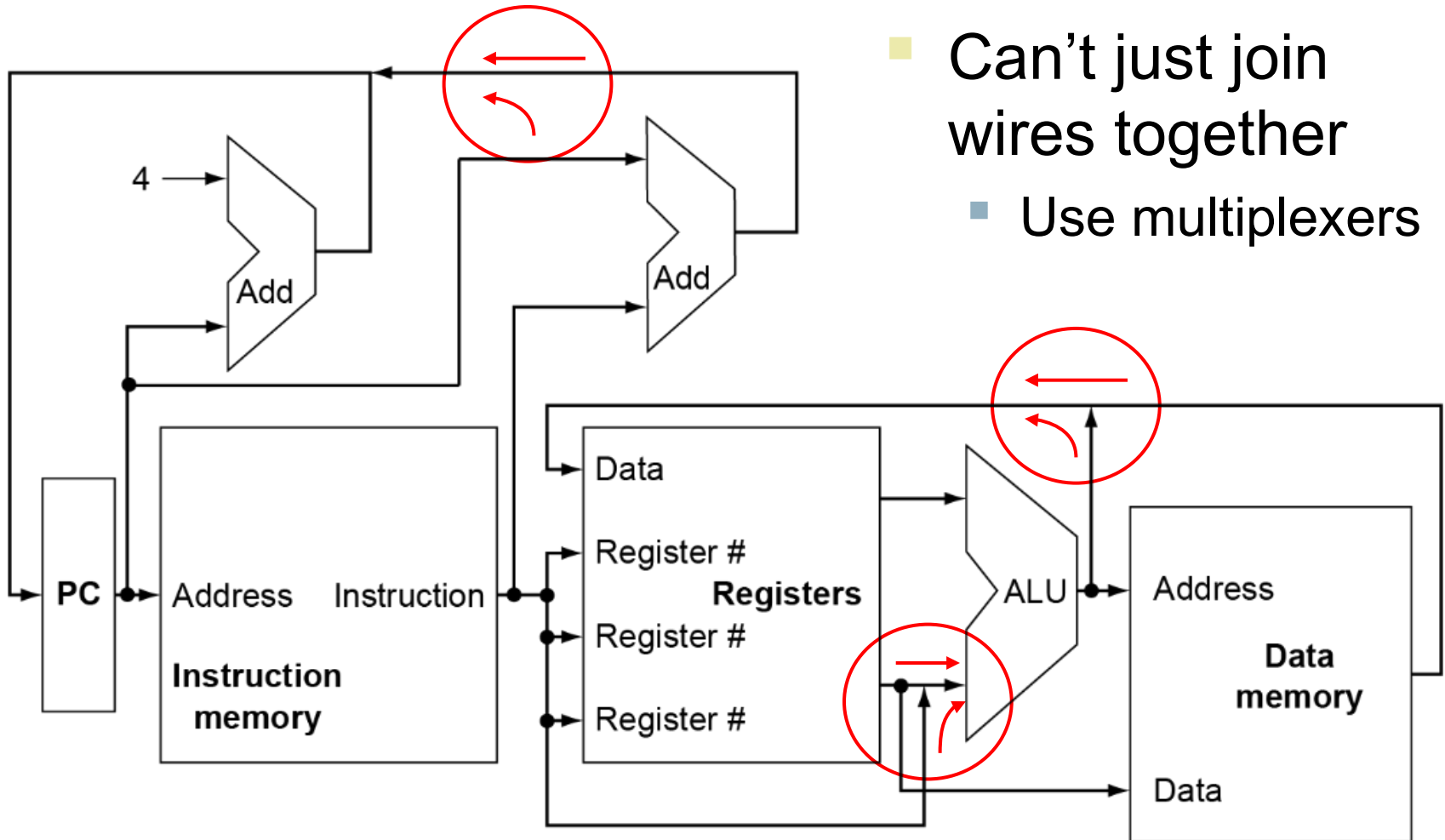


Chapter 4

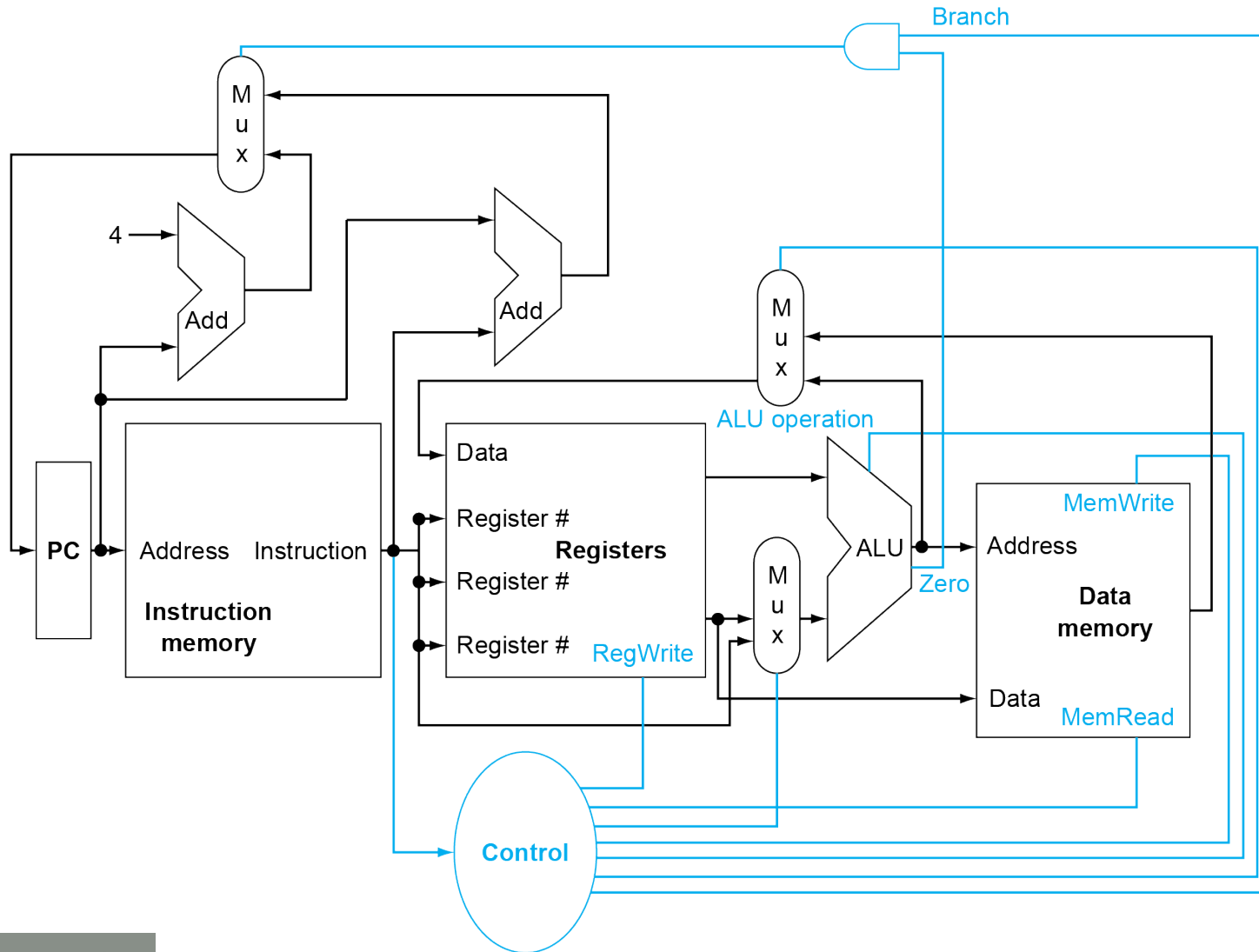
The Processor

Review

- Can't just join wires together
 - Use multiplexers

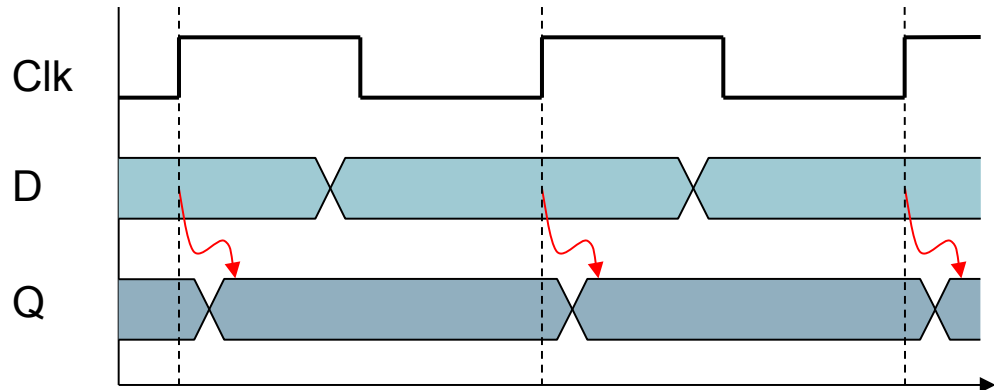
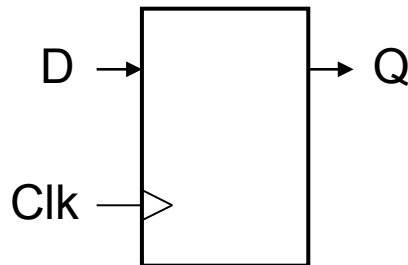


Review



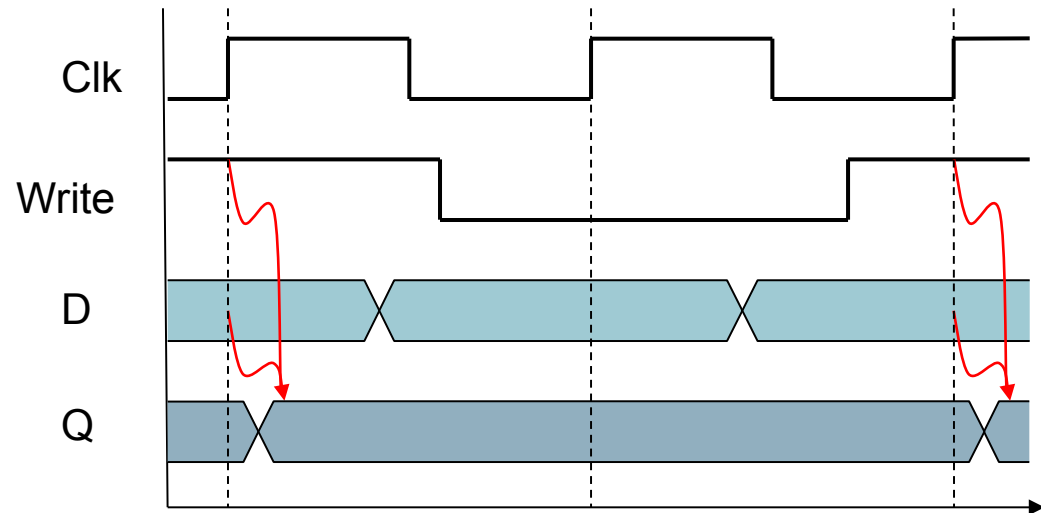
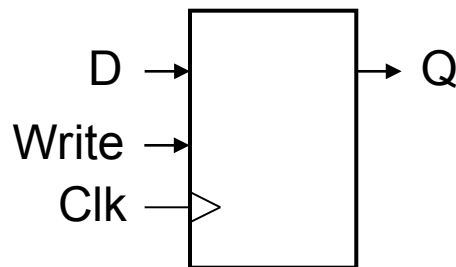
Sequential Elements

- Register: stores data in a circuit
 - Uses a clock signal to determine when to update the stored value
 - Edge-triggered: update when Clk changes from 0 to 1



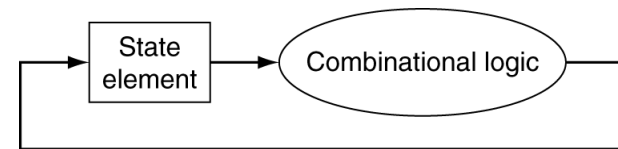
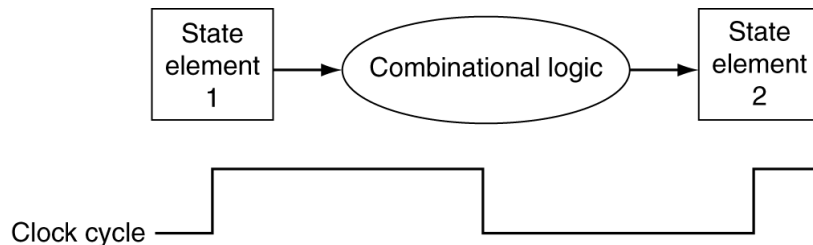
Sequential Elements

- Register with write control
 - Only updates on clock edge when write control input is 1
 - Used when stored value is required later



Clocking Methodology

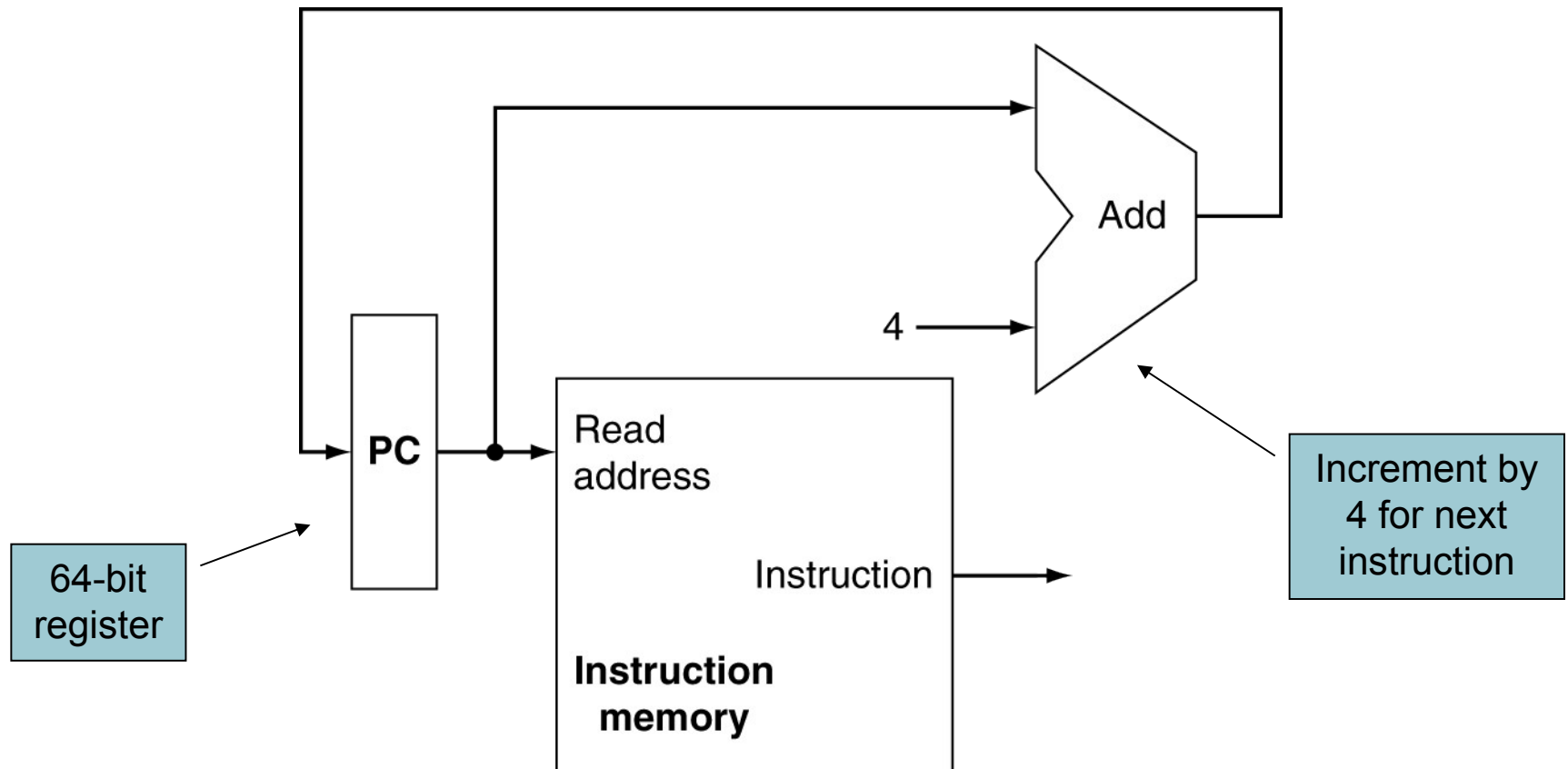
- Combinational logic transforms data during clock cycles
 - Between clock edges
 - Input from state elements, output to state element
 - Longest delay determines clock period



Building a Datapath

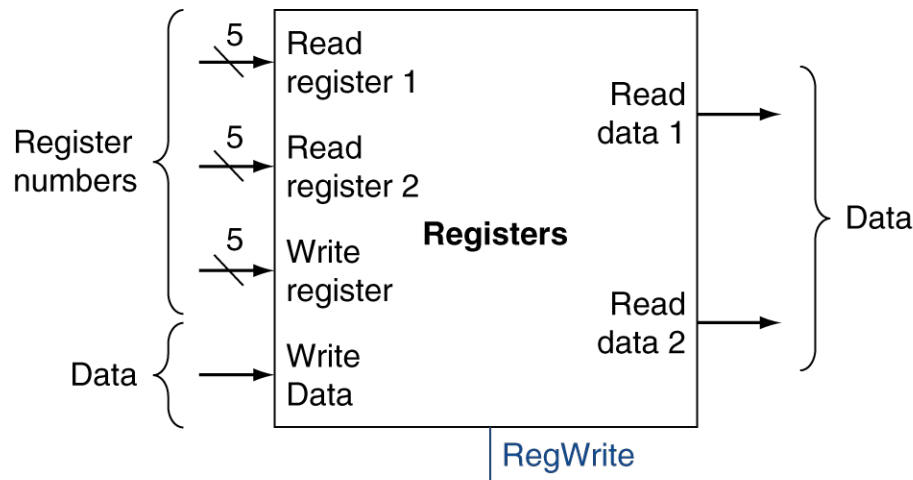
- Datapath
 - Elements that process data and addresses in the CPU
 - Registers, ALUs, mux's, memories, ...
- We will build a RISC-V datapath incrementally
 - Refining the overview design

Instruction Fetch

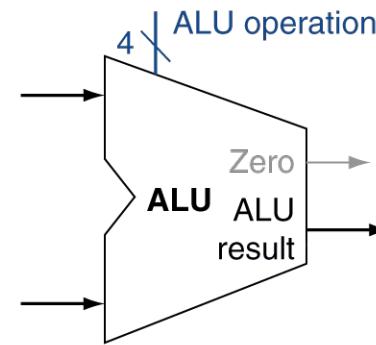


R-Format Instructions

- Read two register operands
- Perform arithmetic/logical operation
- Write register result



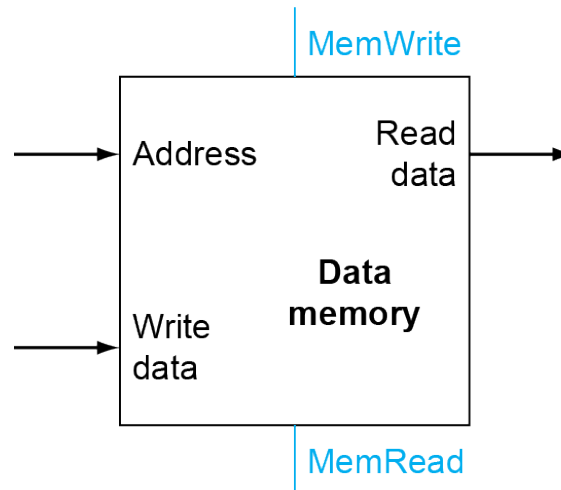
a. Registers



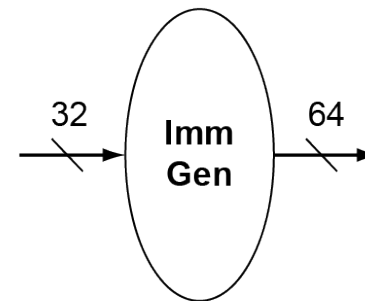
b. ALU

Load/Store Instructions

- Read register operands
- Calculate address using 12-bit offset
 - Use ALU, but sign-extend offset
- Load: Read memory and update register
- Store: Write register value to memory



a. Data memory unit

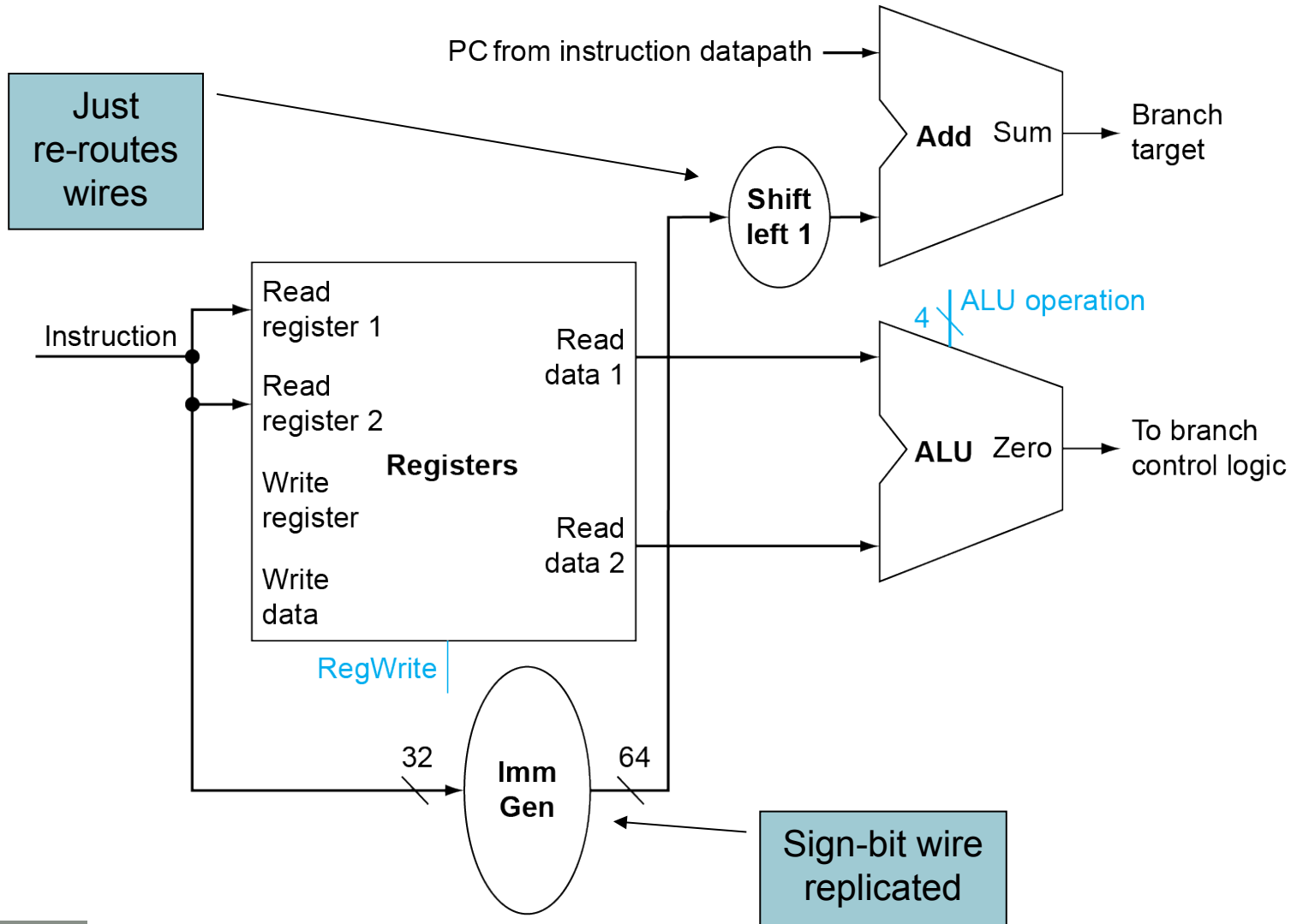


b. Immediate generation unit

Branch Instructions

- Read register operands
- Compare operands
 - Use ALU, subtract and check Zero output
- Calculate target address
 - Sign-extend displacement
 - Shift left 1 place (halfword displacement)
 - Add to PC value

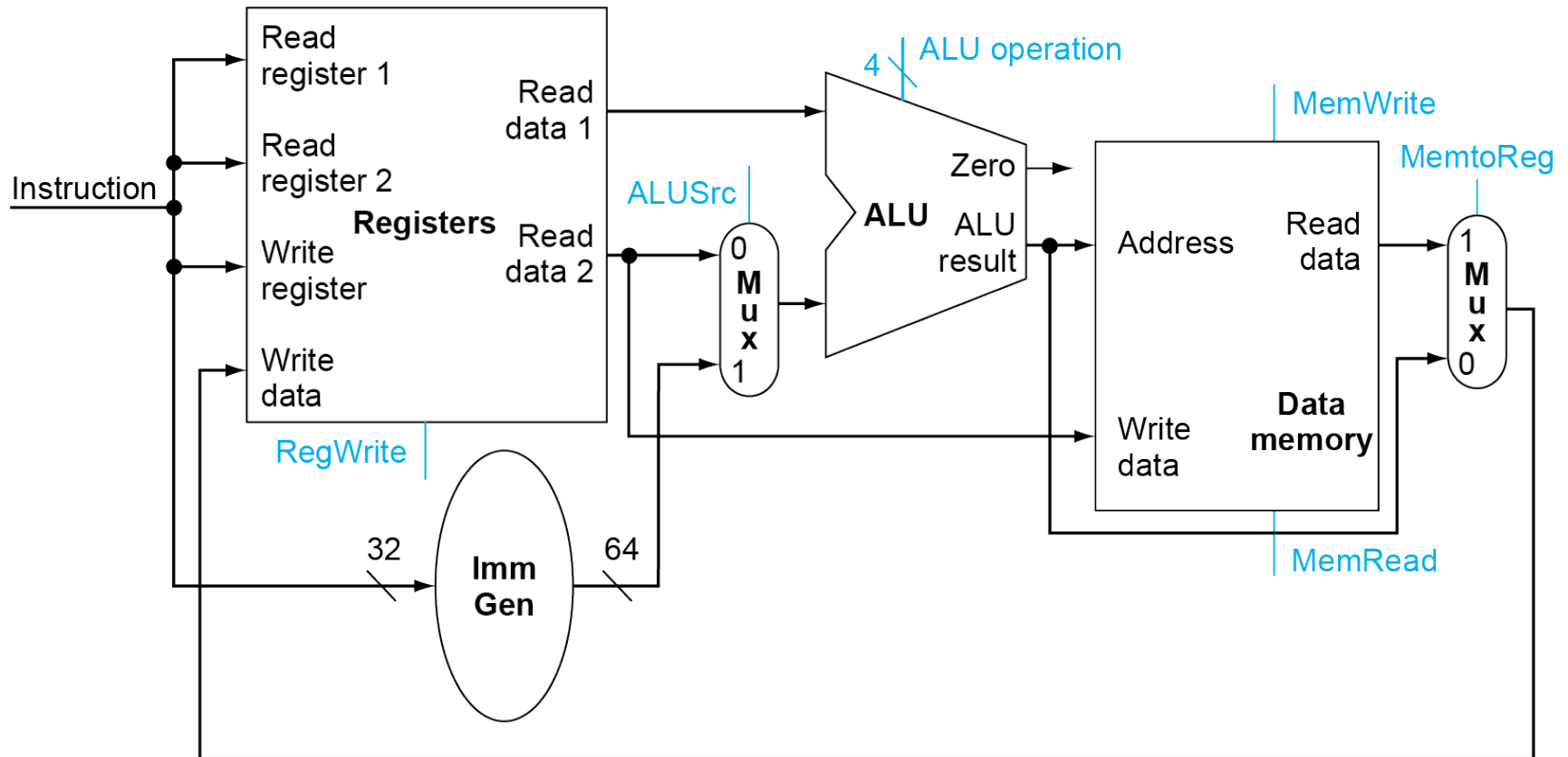
Branch Instructions



Composing the Elements

- First-cut data path does an instruction in one clock cycle
 - Each datapath element can only do one function at a time
 - Hence, we need separate instruction and data memories
- Use multiplexers where alternate data sources are used for different instructions

R-Type/Load/Store Datapath



Full Datapath

